1. Five cryptographic-assumption boundaries in Lean

Review opened 2026-11-01; correction round completed 2026-11-02.

Scoped result. Five finite countermodels/controls and their repaired premises are present in Lean. Five separately assigned semantic reviewers attest `holds`; four first-round audit findings were corrected and re-reviewed. Corrected head `0e096a9d98` passed the exact Lean 4.35 full workspace build, all-target named-declaration verifier, project axiom audit, forbidden-construct checks, and dependency-pin checks (Actions run `41365419446`).

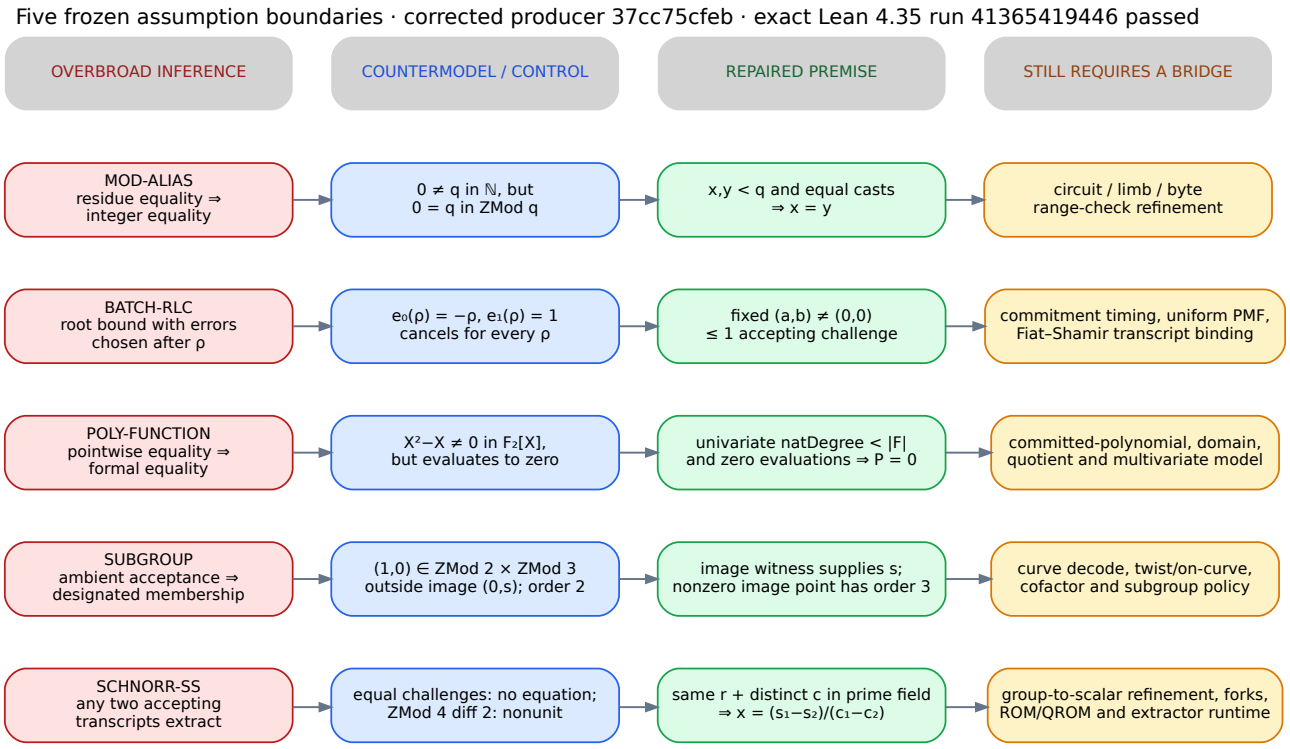


Figure 1: Each red inference is too broad; blue is the finite control, green is the repaired mathematical premise, and amber is the implementation or security bridge that remains unproved.

1.1. What Lean establishes

Joint	Countermodel / control	Repair
`MOD-ALIAS`	Distinct naturals `0,q` alias in `ZMod q`.	Both natural representatives are `< q`; sharp boundary is `0 < q`.
`BATCH-RLC`	`e0(rho)=-rho, e1(rho)=1` cancel for every challenge.	A fixed nonzero coefficient pair accepts at most one field challenge.
`POLY-FUNCTION`	`X^2-X != 0` over `F_2`, yet evaluates to zero.	Univariate `natDegree P < |F|` plus pointwise zero implies `P=0`.
`SUBGROUP`	`(1,0)` lies outside the image `(0,s)` and has order two.	Image membership supplies a scalar; nonzero image points have order three.
`SCHNORR-SS`	Equal challenges give no extraction; a nonzero composite-ring difference may be a nonunit.	Same commitment, distinct challenges, and prime scalar field give the quotient extractor.

The theorem types do not mention a circuit, curve decoder, commitment scheme, random oracle, extractor runtime, or security game. Consequently no named implementation vulnerability, ECDLP attack, or hardness conclusion follows without a separate checked refinement.

1.2. Evidence and adversarial review

The initial producer is `f5a8f8ac8c`; corrected producer `37cc75cfebb`; correction plan `0e096a9d98`. Specs were not edited. The workspace pins Lean `v4.35.0` and Mathlib `cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`.

The 4.34.1 compatibility proxy elaborated the five final modules. Final focused target/verifier tests pass 340/340; all five specs validate; the generated root is current; forbidden-construct scans are clean. The proxy is explicitly not the pinned machine receipt. The exact run's formal-workspace job and receipt-upload step both completed successfully; the accompanying machine record binds run, job, artifact, source hashes, and trust limits.

Five separate reviewers first derived their assigned mathematics from the frozen spec and matching source subsection, then inspected only that proof. All return `holds`. Their attacks exposed and the correction round resolved:

- non-tight modular scope (`2 <= q` versus the sharp `0 < q`);
- batching and polynomial repairs adjacent to, not inside, the audited name;
- a composite-ring Schnorr control that had not stated `0 != 2` explicitly.

Procedure limit: reviewers were separate collaboration sessions, not durable `TASK-*` handoffs; the Markdown plan is not accepted by the mechanical independence checker; all sessions use one model family. This packet changes no official ledger, hypothesis, experiment, or goal status.

1.3. JKSST and hardness boundaries

This tranche does not supply an exact Mordell–Weil height-ball count from rank/radius/minimum, prove the selected-lift height/discriminant comparison, prove Lang's conjecture, or turn "approximately `(2 C2)^r/p`" into a theorem. Those need distinct statements and interfaces. A marginal one-shot bound also does not imply independent or geometric repetition.

DLP, CDH, q-SDH, ROM, AGM, and QROM are open computational assumptions or model-relative statements, not unconditional propositions refuted by these finite controls. The rigorous skeptical task is to formalize their games, reductions, encodings, and loss factors.

1.4. Recommended next five

1. `ONE-SHOT-HISTORY`: correlated-trial countermodels; repair with a uniform conditional-hazard recurrence. This is the direct JKSST priority.
2. `CUBIC-EVAL-RANK`: ten collinear points do not give ten independent cubic conditions; repair with a full-rank evaluation-matrix premise.
3. `COFACTOR-PUSHFORWARD`: exact image/fiber/uniform-pushforward counts under a proved group decomposition and scalar actions.
4. `FS-ZERO-HASH`: constant-zero Fiat–Shamir countermodel, explicitly short of a ROM/QROM forking theorem.
5. `PEDERSEN-RELATION-KERNEL`: alternate openings from a known generator relation; computational binding remains an assumption/game theorem.

Full statements, hashes, review reports, missing bridges, and links to proof sources are in the accompanying `README.md`.